Analytical Chemistry Cumulative Exam
April 3, 2009
Basic Electrochemical Concepts

1. (10 points) In electrochemistry, what is the double layer? Where is it? Draw a diagram and label and name each of the regions and describe what is present in each region. What component in electronics best represents the behavior of the double layer when a potential is applied across it?

2. (12 points) What is a reference electrode? How does it participate in an electrochemical experiment (such as in cyclic voltammetry)? Give one example of the chemistry and composition involved in a commonly used reference electrode and write the half reaction. Address each substance in the half reaction and consider whether a change in the amount of that substance would cause the reference potential to change.

3. (10 points) What is a counter or auxiliary electrode and what does it do in an electrochemical experiment (such as in cyclic voltammetry)? Name some considerations in selecting an appropriate material for a counter electrode.

4. (8 points) Give at least two reasons why it is a good idea to have electrolyte in the sample solution when carrying out electrochemical measurements.

3. (10 points) The standard electrode potential, E°, for Ni^{2+} reduction to Ni° vs. reference electrode #1 in the literature is -0.257 V. But you have a different reference electrode, reference electrode #2. What is E° for Ni^{2+} reduction vs. your reference electrode #2? (Note that the standard electrode potential of reference electrode #2 vs. reference electrode #1 is +0.197 V.)

4. (20 points) Describe the cyclic voltammetry (CV) experiment. What information does the electrochemical response give you? (Name 3 measurements that you can make from the voltammogram and explain the kind of information about the chemistry that they provide. Try to be as quantitative as possible. Write equations, if possible.) Draw an example of the CV electrochemical response and label the features of the response that are measured. Assume that the electrode is a macroelectrode and that the species in solution that you are analyzing begins in its oxidized for, "O", and has fast and reversible electron transfer kinetics to form the reduced form "R". Assume that the E° value is 0.00 V versus your reference electrode and that the range of the applied potentials in the CV experiment include the E° value.

5. (10 points) What would happen to the cyclic voltammogram as you slow down the kinetics of electron transfer of O to R in #4, but keep the scan rate and all other conditions the same? (Draw the outcome.)

6. (10 points) How would the cyclic voltammogram response in #4 change if you were to swap the macroelectrode to a microelectrode so that the diffusion length is much larger than the smallest dimension of the electrode? Draw the corresponding response.

7. (10 points) What happens to the cyclic voltammogram in #6 as you increase the scan rate so that the diffusion length becomes much shorter than the smallest dimension of the electrode?